DAY_2

- ❖ **Introduction to VPNsTheory :Types of VPNs: Site-to-site, remote access.Practical :Configure a basic site-to-site VPN in Packet Tracer.Task: Submit configuration steps for a VPN.**

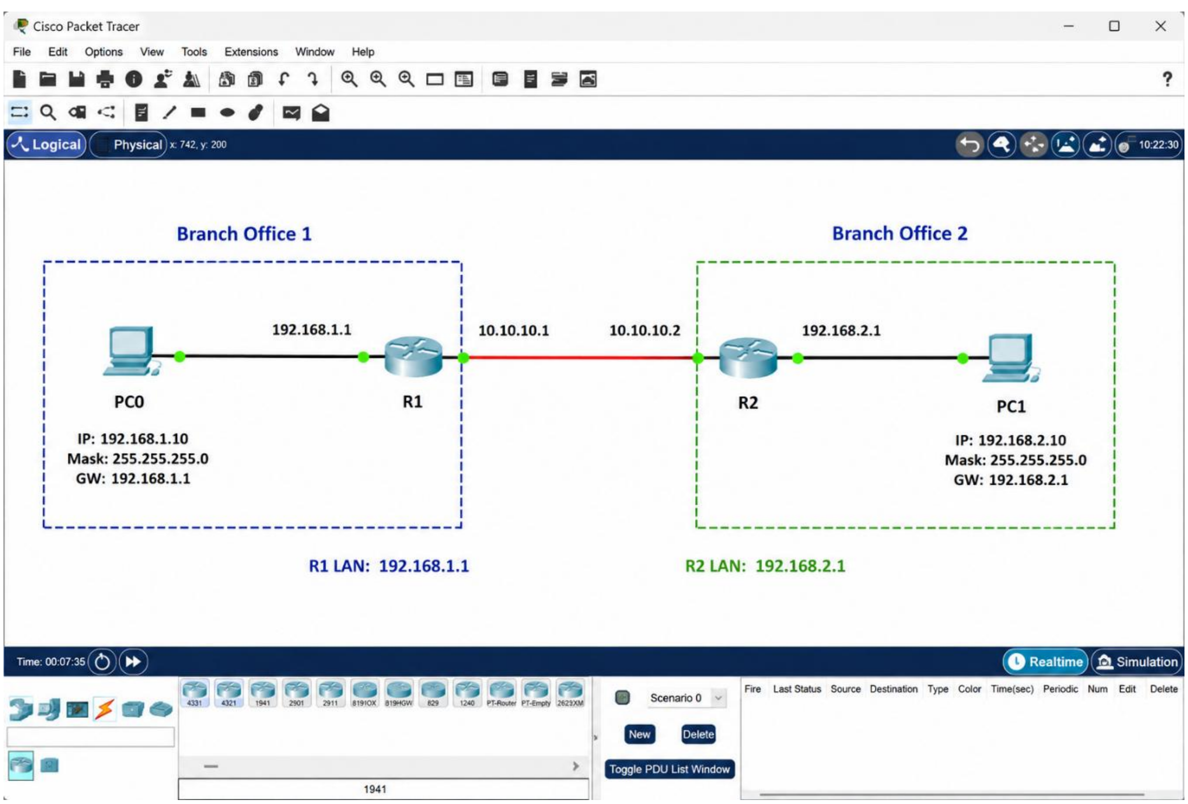
1. In Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.

ANS :

- ➢ A **VPN (Virtual Private Network)** creates a secure logical connection between devices or networks over another network, such as the Internet. It allows users or branch offices to communicate privately even when they are geographically separated.

▪ Types of VPNs

- ✓ **Site-to-Site VPN –** Connects two complete networks, such as two branch offices.
- ✓ **Remote Access VPN –** Connects an individual user or device remotely to a private network.



- **Router0**

```
enable
configure terminal

interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 10.10.10.1 255.255.255.252
no shutdown
    exit
```

- **Router1**

```
enable
configure terminal

interface gigabitEthernet0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 10.10.10.2 255.255.255.252
no shutdown
    exit
```

2. Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch. Hint: Use the 'interface tunnel' command on both routers and set the tunnel source and destination to the routers' public-facing interfaces.

ANS :

- **Router0 – GRE Tunnel Configuration**

```
enable
configure terminal
```

```
interface tunnel0
ip address 172.16.1.1 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.10.2
no shutdown
    exit
```

- **Router1 – GRE Tunnel Configuration**

```
enable
configure terminal

interface tunnel0
ip address 172.16.1.2 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.10.1
no shutdown
    exit
```

- **Verify the GRE Tunnel**

  - On Router0:
    show ip interface brief

  - You should see:
    Tunnel0 172.16.1.1 up up

  - On Router1:
    show ip interface brief

  - You should see:
    Tunnel0 172.16.1.2 up up

  - You can also test the tunnel:

    From Router0:
        ping 172.16.1.2
    From Router1:
        ping 172.16.1.1

3. Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature. Hint: Use the VPN tab on the PC and set the correct server IP, username, and password.

ANS :

- **General Packet Tracer Steps**

1) Open the VPN client PC.
2) Go to the **Desktop** section.
3) Open the available **VPN/VPN Client configuration** feature.
4) Enter the VPN server address, for example:
   - Server IP: 10.10.10.1
   - Username: vpnuser
   - Password: cisco123
5) Configure the corresponding username and password on the VPN server/router according to the Packet Tracer device and VPN feature being used.
6) Click **Connect**.
7) After the VPN connection is established, test connectivity to internal resources using `ping`.

4. List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.

ANS :

Feature	**Site-to-Site VPN**	**Remote Access VPN**
Connection	Connects two networks	Connects an individual user/device
VPN Endpoints	Usually routers/firewalls	User device and VPN server
Users	Entire branch network	Individual remote users
Configuration	Configured mainly on network devices	Requires client login
Authentication	Device-to-device	Username/password or certificates
Example	Ahmedabad branch to Mumbai branch	Employee working from home

- **Site-to-Site VPN Advantages**

1. **Connects complete networks automatically** – Users at both offices can communicate without individually starting a VPN.
2. **Centralized management** – VPN configuration is managed mainly on routers or firewalls.

- **Site-to-Site VPN Disadvantages**

  1. **More complex initial configuration** – Routing, tunnel endpoints, and security settings must be correctly configured.
  2. **Less suitable for individual users** – It is designed primarily for connecting networks rather than individual remote employees.

- **Remote Access VPN Advantages**

  1. **Flexible** – Users can securely connect from home, hotels, or other remote locations.
  2. **Individual authentication** – Each user can have their own username and password.

- **Remote Access VPN Disadvantages**

  1. **Requires client configuration** – Each remote user must configure and connect using a VPN client.
  2. **Management can become difficult** – Managing many users, passwords, and permissions requires additional administration.

5. Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.

ANS :

- **Router0 Complete Configuration**

```
enable
configure terminal

! Configure Branch 1 LAN
interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

! Configure public-facing/WAN interface
interface gigabitEthernet0/1
ip address 10.10.10.1 255.255.255.252
no shutdown
exit

! Configure GRE Tunnel
interface tunnel0
ip address 172.16.1.1 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.10.2
no shutdown
```

```
exit

! Route to Branch 2 through the GRE tunnel
ip route 192.168.2.0 255.255.255.0 172.16.1.2

end
    write memory
```

- **Router1 Complete Configuration**

```
enable
configure terminal

! Configure Branch 2 LAN
interface gigabitEthernet0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

! Configure public-facing/WAN interface
interface gigabitEthernet0/1
ip address 10.10.10.2 255.255.255.252
no shutdown
exit

! Configure GRE Tunnel
interface tunnel0
ip address 172.16.1.2 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.10.1
no shutdown
exit

! Route to Branch 1 through the GRE tunnel
ip route 192.168.1.0 255.255.255.0 172.16.1.1

end
    write memory
```